Toolchain Options === ABI

- When building a toolchain, the ABI used to generate binaries needs to be defined
- ABI, for *Application Binary Interface*, defines the calling conventions (how function arguments are passed, how the return value is passed, how system calls are made) and the organization of structures (alignment, etc.)
- All binaries in a system are typically compiled with the same ABI, and the kernel must understand this ABI.
- On ARM 32-bit, two main ABIs: *EABI* and *EABIhf*
  - *EABIhf* passes floating-point arguments in floating-point registers $arrow.r$_needs an ARM processor with a FPU
- On RISC-V, several ABIs: *ilp32*, *ilp32f*, *ilp32d*, *lp64*, *lp64f*, and *lp64d*

- https://en.wikipedia.org/wiki/Application_Binary_Interface

Floating point support

- All ARMv7-A (32-bit) and ARMv8-A (64-bit) processors have a floating point unit
- RISC-V cores with the F extension have a floating point unit
- Some older ARM cores (ARMv4/ARMv5) or some RISC-V cores may not have a floating point unit
- For processors without a floating point unit, two solutions for floating point computation:
  - Generate *hard float code* and rely on the kernel to emulate the floating point instructions. This is very slow.
  - Generate *soft float code*, so that instead of generating floating point instructions, calls to a user space library are generated
- Decision taken at toolchain configuration time
- For processors with a floating point unit, sometimes different FPU are possible. For example on ARM: VFPv3, VFPv3-D16, VFPv4, VFPv4-D16, etc.

CPU optimization flags

- GNU tools (gcc, binutils) can only be compiled for a specific target architecture at a time (ARM, x86, RISC-V…)
- gcc offers further options:
  - `-march` allows to select a specific target instruction set
  - `-mtune` allows to optimize code for a specific CPU
  - For example: `-march=armv7 -mtune=cortex-a8`
  - `-mcpu=cortex-a8` can be used instead to allow gcc to infer the target instruction set (`-march=armv7`) and cpu optimizations (`-mtune=cortex-a8`)
  - https://gcc.gnu.org/onlinedocs/gcc/ARM-Options.html
- At the GNU toolchain compilation time, values can be chosen. They are used:
  - As the default values for the cross-compiling tools, when no other `-march`, `-mtune`, `-mcpu` options are passed
  - To compile the C library
- Note: LLVM (Clang, LLD…) utilities support multiple target architectures at once.